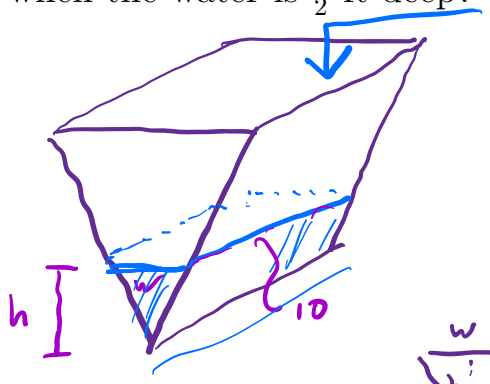
Problem 5. WORKED IN CLASS

A trough 10 ft long has ends shaped like isosceles triangles, 3 ft across the top and 1 ft deep. Water pours in at 12 ft^3/min. How fast is the depth rising when the water is $\frac{1}{2}$ ft deep?

1. Draw + Label
2. What & when
3. Equation
4. Differentiate w.r.t. t
5. Sub + Solve

(1)



(2) $\frac{dh}{dt} = ?$ when $h = \frac{1}{2}$

$\frac{dV}{dt} = 12$



(3) $V = (\text{cross-sectional area}) \cdot L = \left(\frac{1}{2}wh\right) \cdot 10$

$V = 10 \cdot \frac{1}{2} wh = 5wh$

$V = 5wh = 5 \cdot 3h \, h = 15h^2$

Similar Δ's

$\frac{w}{h} = \frac{3}{1}$

$w = 3h$

(4) $\frac{d}{dt}V = \frac{d}{dt} 15h^2$

$\frac{dV}{dt} = 15 \cdot 2h \frac{dh}{dt} = 30h \frac{dh}{dt}$

(5) $12 = 30 \cdot \frac{1}{2} \frac{dh}{dt}$

$12 = 15 \frac{dh}{dt}$

$\frac{dh}{dt} = \frac{12}{15} = \frac{\cancel{3} \cdot 4}{\cancel{3} \cdot 5} \frac{ft}{min} = .8 \frac{ft}{min}$